

Homework 3: Classification

IE 2064 Data Science

Dataset choice

For this assignment, we will be using the Hepatic Injury dataset. This dataset was taken from Applied Predictive Models (APM) Chapter 1.4.

Simplify injury

Below are the levels of injury after being simplified:

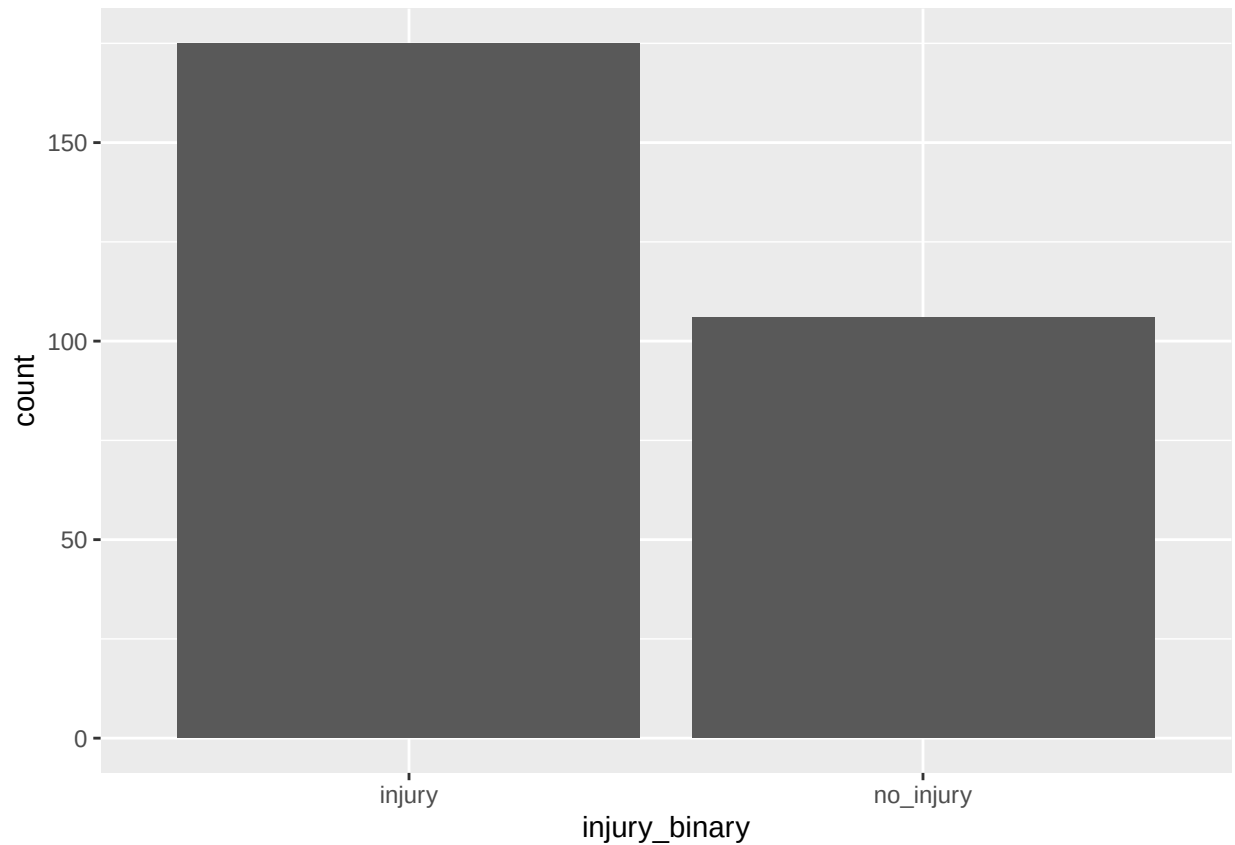
```
## [1] injury    no_injury
## Levels: injury no_injury
```

Identify an outcome variable to be predicted

- **injury_binary** (categorical) = If the liver was damaged when exposed to each compound. A summary of this column is:

```
##      injury no_injury
##      175      106
```

Below is a histogram of injury_binary



Preprocessing

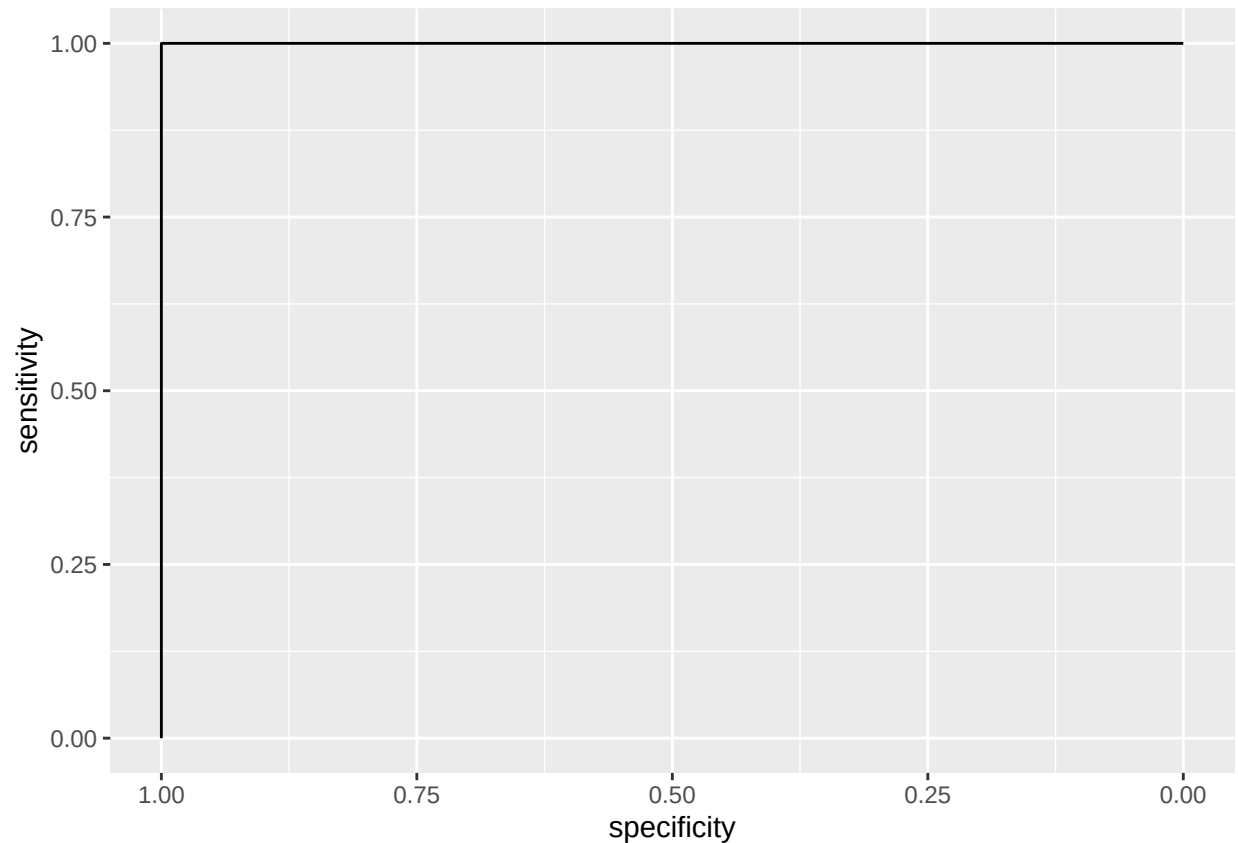
Since there are no missing values in the data set, the only pre-processing that needed to be done was removing predictors with zero variance. Within each train statement, appropriate pre-processing will be applied.

Logistic Regression

```
## Generalized Linear Model
##
## 281 samples
## 103 predictors
## 2 classes: 'injury', 'no_injury'
##
## Pre-processing: centered (104), scaled (104)
## Resampling: Cross-Validated (10 fold)
## Summary of sample sizes: 252, 252, 253, 253, 254, 253, ...
## Resampling results:
##
## Accuracy Kappa
## 1 1

## Setting levels: control = no_injury, case = injury

## Setting direction: controls < cases
```



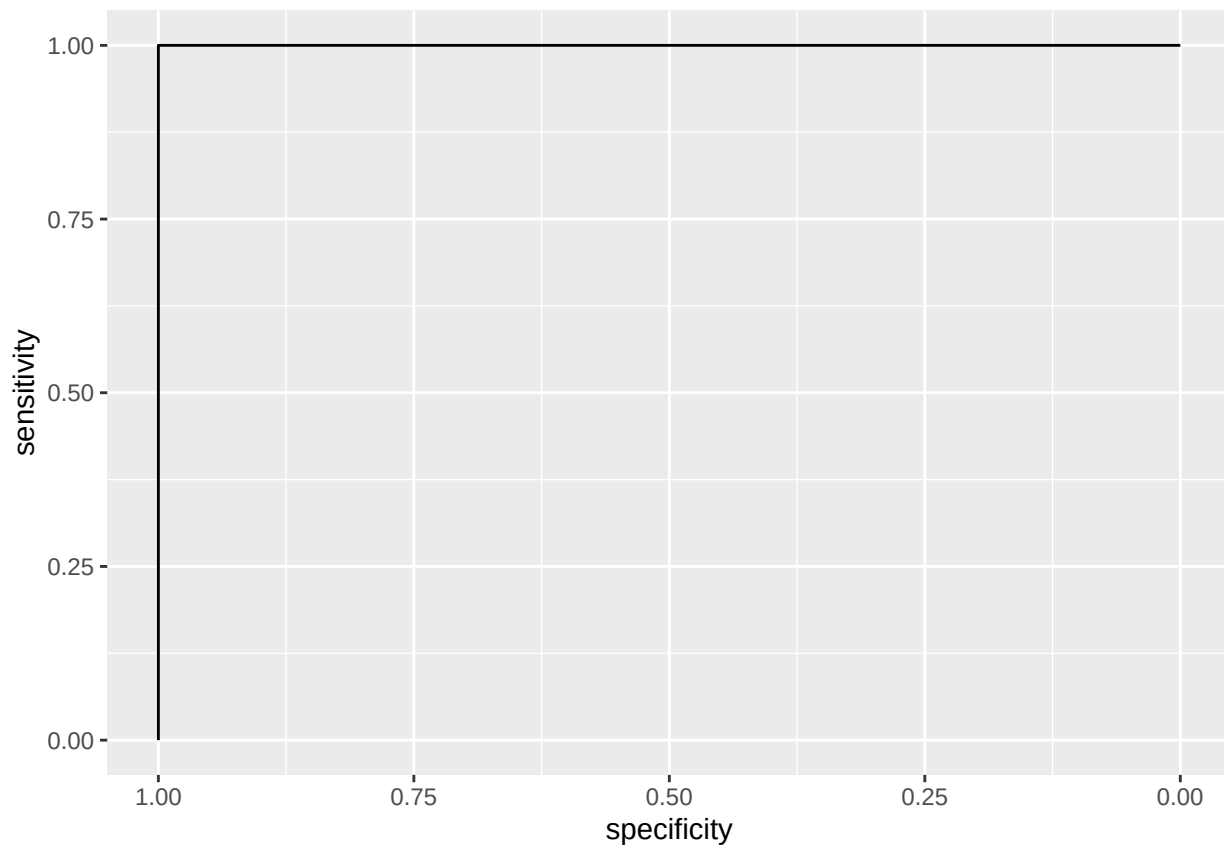
```
## Area under the curve: 1
```

Linear discriminant analysis

```
## Linear Discriminant Analysis
##
## 281 samples
## 103 predictors
## 2 classes: 'injury', 'no_injury'
##
## Pre-processing: centered (104), scaled (104), principal component
## signal extraction (104)
## Resampling: Cross-Validated (10 fold)
## Summary of sample sizes: 253, 252, 252, 254, 253, 253, ...
## Resampling results:
##
## Accuracy  Kappa
## 0.9964286 0.9923913

## Setting levels: control = no_injury, case = injury

## Setting direction: controls < cases
```



Area under the curve: 1

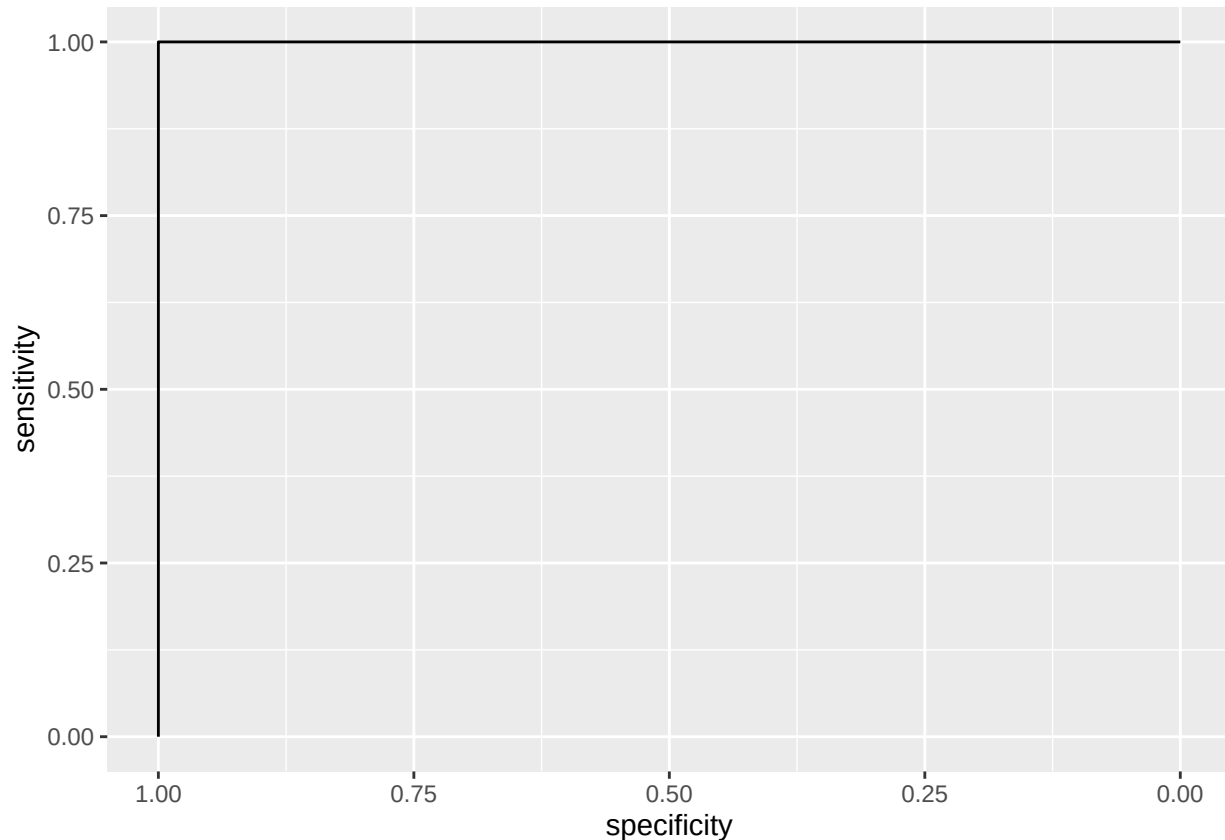
Penalized models

##	alpha	lambda	Accuracy	Kappa	AccuracySD	KappaSD
## 1	0.00	0.000	0.9890212	0.9761751	0.01768127	0.03838583
## 2	0.00	0.005	0.9890212	0.9761751	0.01768127	0.03838583
## 3	0.00	0.010	0.9890212	0.9761751	0.01768127	0.03838583
## 4	0.00	0.015	0.9890212	0.9761751	0.01768127	0.03838583
## 5	0.00	0.020	0.9890212	0.9761751	0.01768127	0.03838583
## 6	0.25	0.000	1.0000000	1.0000000	0.00000000	0.00000000
## 7	0.25	0.005	1.0000000	1.0000000	0.00000000	0.00000000
## 8	0.25	0.010	1.0000000	1.0000000	0.00000000	0.00000000
## 9	0.25	0.015	1.0000000	1.0000000	0.00000000	0.00000000
## 10	0.25	0.020	1.0000000	1.0000000	0.00000000	0.00000000
## 11	0.50	0.000	1.0000000	1.0000000	0.00000000	0.00000000
## 12	0.50	0.005	1.0000000	1.0000000	0.00000000	0.00000000
## 13	0.50	0.010	1.0000000	1.0000000	0.00000000	0.00000000
## 14	0.50	0.015	1.0000000	1.0000000	0.00000000	0.00000000
## 15	0.50	0.020	1.0000000	1.0000000	0.00000000	0.00000000
## 16	0.75	0.000	1.0000000	1.0000000	0.00000000	0.00000000
## 17	0.75	0.005	1.0000000	1.0000000	0.00000000	0.00000000
## 18	0.75	0.010	1.0000000	1.0000000	0.00000000	0.00000000
## 19	0.75	0.015	1.0000000	1.0000000	0.00000000	0.00000000
## 20	0.75	0.020	1.0000000	1.0000000	0.00000000	0.00000000

```
## 21  1.00  0.000  1.00000000  1.00000000  0.00000000  0.00000000
## 22  1.00  0.005  1.00000000  1.00000000  0.00000000  0.00000000
## 23  1.00  0.010  1.00000000  1.00000000  0.00000000  0.00000000
## 24  1.00  0.015  1.00000000  1.00000000  0.00000000  0.00000000
## 25  1.00  0.020  1.00000000  1.00000000  0.00000000  0.00000000
```

```
## Setting levels: control = no_injury, case = injury
```

```
## Setting direction: controls < cases
```



```
## Area under the curve: 1
```

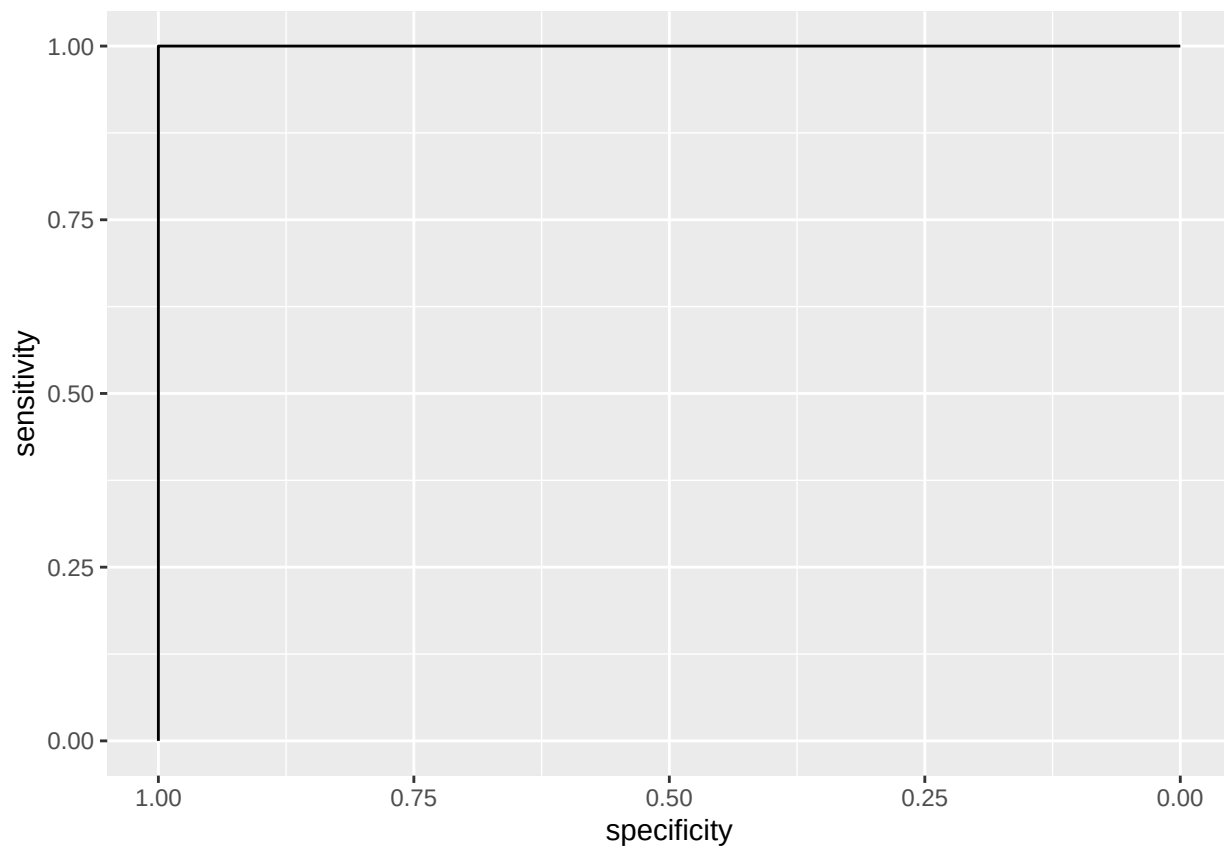
Classification trees

```
## CART
##
## 281 samples
## 103 predictors
## 2 classes: 'injury', 'no_injury'
##
## No pre-processing
## Resampling: Cross-Validated (10 fold)
## Summary of sample sizes: 252, 252, 254, 252, 253, 253, ...
## Resampling results across tuning parameters:
```

```
##
##   cp   Accuracy  Kappa
## 0.000 1         1
## 0.005 1         1
## 0.010 1         1
## 0.015 1         1
## 0.020 1         1
## 0.030 1         1
## 0.040 1         1
##
## Accuracy was used to select the optimal model using the largest value.
## The final value used for the model was cp = 0.04.

## Setting levels: control = no_injury, case = injury

## Setting direction: controls < cases
```



```
## Area under the curve: 1
```

Random forests

```
## Random Forest
##
## 281 samples
```

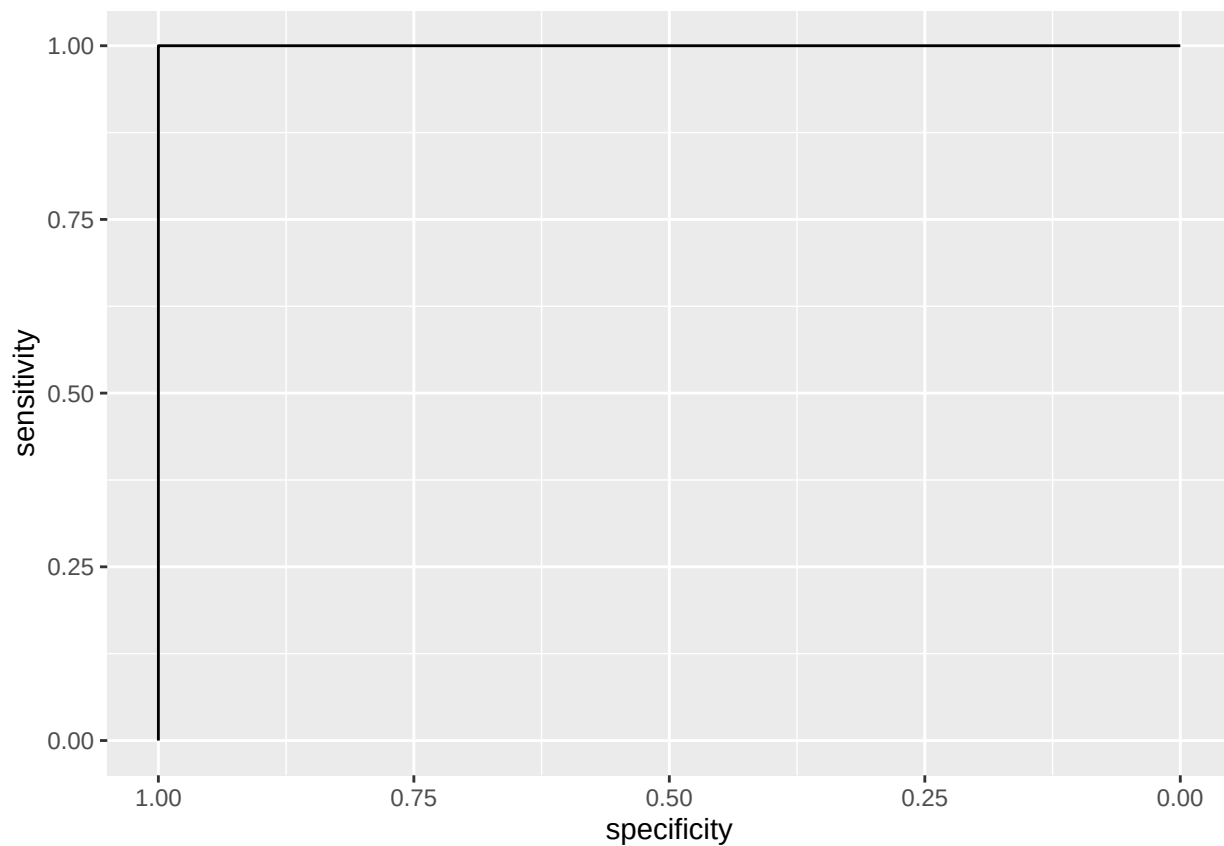
```

## 103 predictors
## 2 classes: 'injury', 'no_injury'
##
## No pre-processing
## Resampling: Cross-Validated (10 fold)
## Summary of sample sizes: 253, 253, 252, 253, 254, 253, ...
## Resampling results across tuning parameters:
##
##  mtry  Accuracy  Kappa
##    3    0.9151387 0.8106613
##    5    0.9822660 0.9622449
##    7    0.9857143 0.9702748
##   10    0.9965517 0.9925450
##   12    1.0000000 1.0000000
##   15    1.0000000 1.0000000
##
## Accuracy was used to select the optimal model using the largest value.
## The final value used for the model was mtry = 12.

## Setting levels: control = no_injury, case = injury

## Setting direction: controls < cases

```



```

## Area under the curve: 1

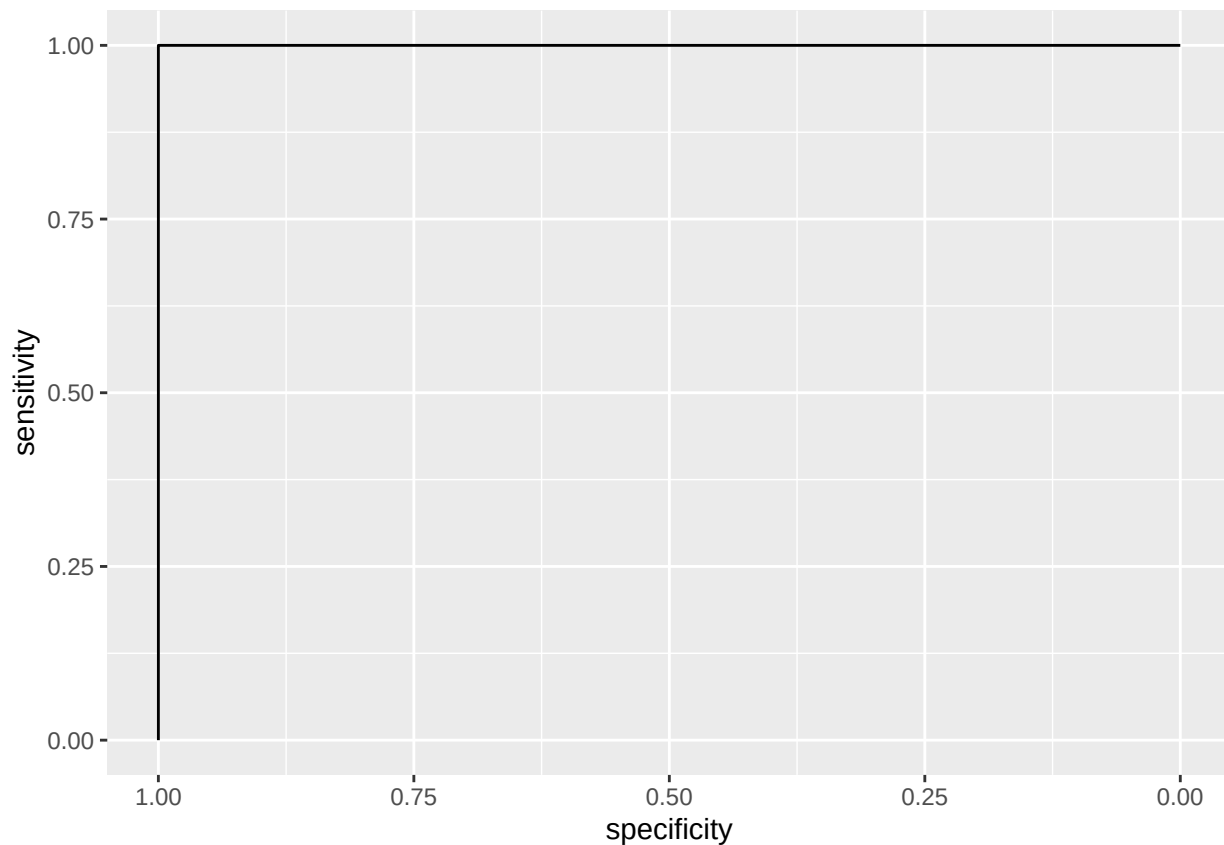
```

Support Vector Machines (SVM)

```
## Support Vector Machines with Linear Kernel
##
## 281 samples
## 103 predictors
## 2 classes: 'injury', 'no_injury'
##
## Pre-processing: centered (104), scaled (104)
## Resampling: Cross-Validated (10 fold)
## Summary of sample sizes: 252, 253, 252, 254, 253, 254, ...
## Resampling results across tuning parameters:
##
## C      Accuracy  Kappa
## 0.10  1          1
## 0.15  1          1
## 0.20  1          1
## 0.25  1          1
## 0.30  1          1
##
## Accuracy was used to select the optimal model using the largest value.
## The final value used for the model was C = 0.1.

## Setting levels: control = no_injury, case = injury

## Setting direction: controls < cases
```



```
## Area under the curve: 1
```

Neural Nets

```
## Warning: executing %dopar% sequentially: no parallel backend registered
```

```
## Model Averaged Neural Network
```

```
##
```

```
## 281 samples
```

```
## 103 predictors
```

```
## 2 classes: 'injury', 'no_injury'
```

```
##
```

```
## Pre-processing: centered (104), scaled (104)
```

```
## Resampling: Cross-Validated (10 fold)
```

```
## Summary of sample sizes: 252, 253, 252, 254, 253, 254, ...
```

```
## Resampling results across tuning parameters:
```

```
##
```

##	decay	size	Accuracy	Kappa
##	0	1	0.9041416	0.80193735
##	0	3	0.9258256	0.84447496
##	0	5	0.9320699	0.85762093
##	1	1	0.9962963	0.99189189
##	1	3	0.9890212	0.97617509
##	1	5	0.9888889	0.97532747
##	10	1	0.6228790	0.00000000
##	10	3	0.6228790	0.00000000
##	10	5	0.6480296	0.08216398

```
##
```

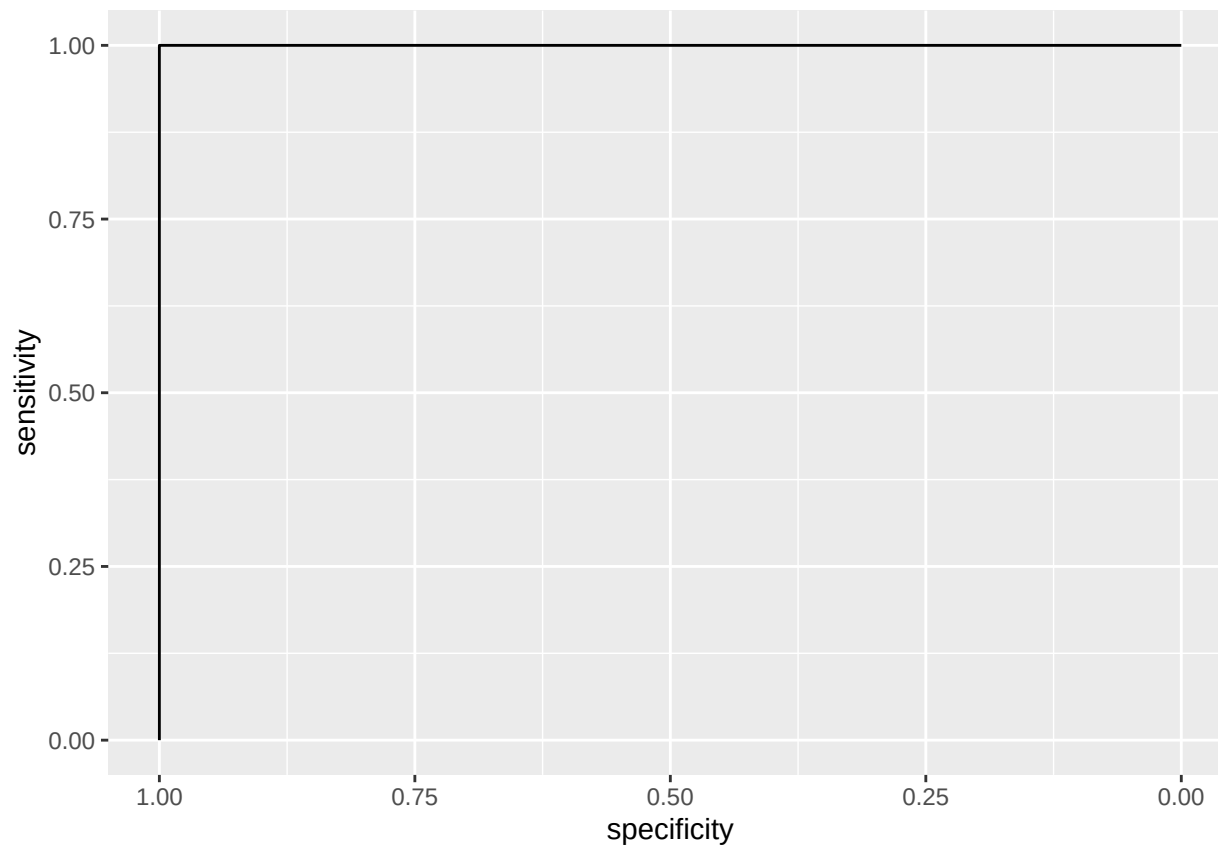
```
## Tuning parameter 'bag' was held constant at a value of TRUE
```

```
## Accuracy was used to select the optimal model using the largest value.
```

```
## The final values used for the model were size = 1, decay = 1 and bag = TRUE.
```

```
## Setting levels: control = no_injury, case = injury
```

```
## Setting direction: controls < cases
```



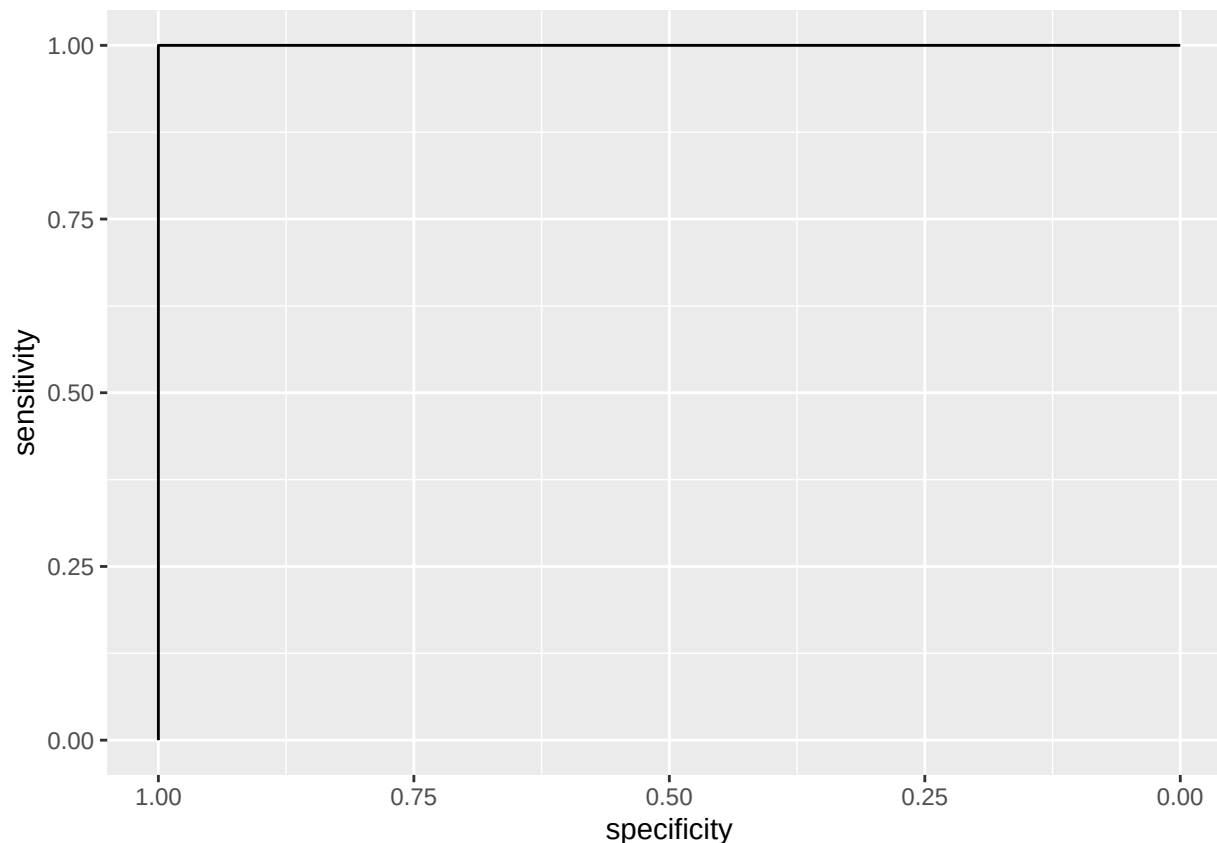
```
## Area under the curve: 1
```

Non-linear discriminant analysis

```
## Quadratic Discriminant Analysis
##
## 281 samples
## 103 predictors
## 2 classes: 'injury', 'no_injury'
##
## Pre-processing: centered (104), scaled (104), principal component
## signal extraction (104)
## Resampling: Cross-Validated (10 fold)
## Summary of sample sizes: 252, 252, 254, 254, 253, 253, ...
## Resampling results:
##
## Accuracy Kappa
## 0.9323344 0.8463958

## Setting levels: control = no_injury, case = injury

## Setting direction: controls < cases
```



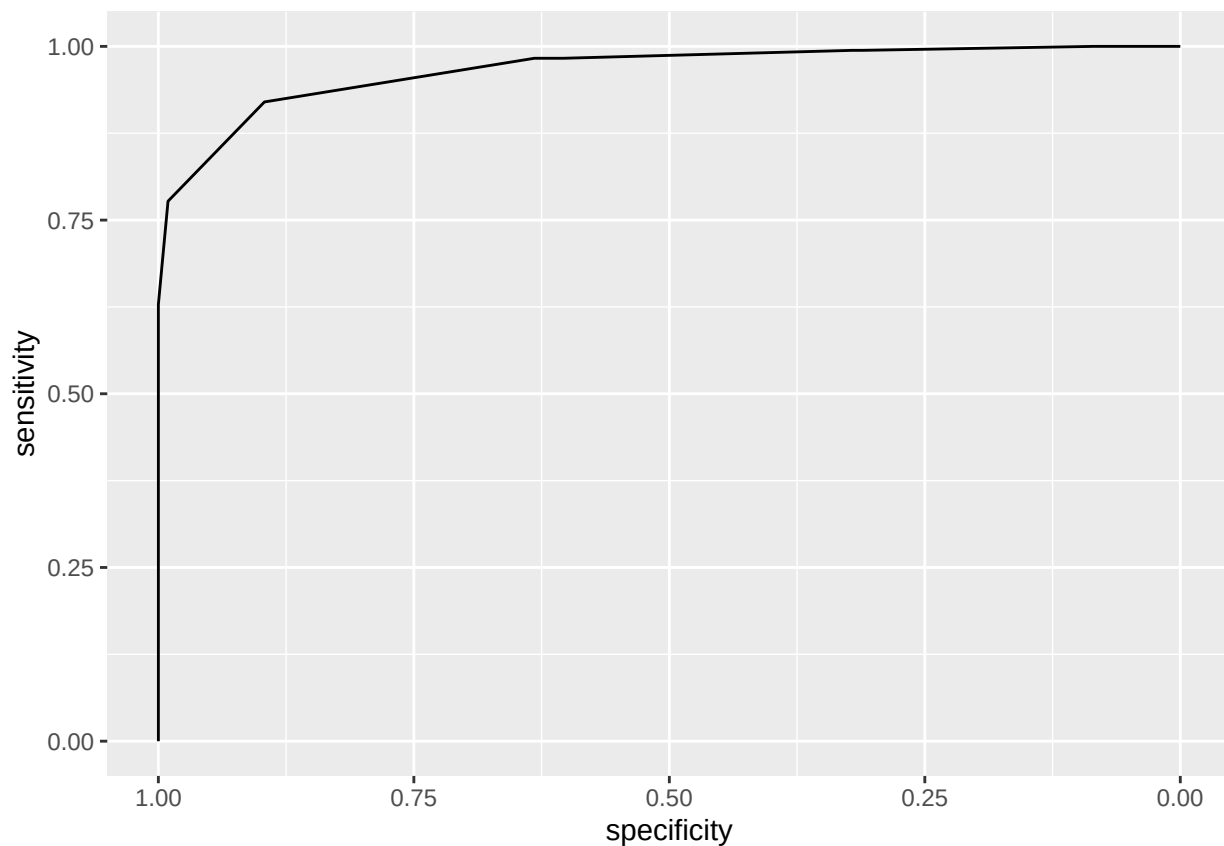
```
## Area under the curve: 1
```

K-nearest Neighbors (KNN)

```
## k-Nearest Neighbors
##
## 281 samples
## 103 predictors
## 2 classes: 'injury', 'no_injury'
##
## Pre-processing: centered (104), scaled (104)
## Resampling: Cross-Validated (10 fold)
## Summary of sample sizes: 252, 253, 252, 254, 253, 254, ...
## Resampling results across tuning parameters:
##
## k Accuracy Kappa
## 2 0.7437557 0.4420790
## 4 0.7784939 0.5082790
## 6 0.7867999 0.5272040
## 8 0.8221447 0.6042947
## 10 0.8185915 0.5844828
## 12 0.7756705 0.4738058
##
## Accuracy was used to select the optimal model using the largest value.
## The final value used for the model was k = 8.
```

```
## Setting levels: control = no_injury, case = injury
```

```
## Setting direction: controls < cases
```



```
## Area under the curve: 0.9657
```

Naive Bayes

```
## Naive Bayes
```

```
##
```

```
## 281 samples
```

```
## 103 predictors
```

```
## 2 classes: 'injury', 'no_injury'
```

```
##
```

```
## No pre-processing
```

```
## Resampling: Cross-Validated (10 fold)
```

```
## Summary of sample sizes: 252, 253, 252, 254, 253, 254, ...
```

```
## Resampling results:
```

```
##
```

```
## Accuracy Kappa
```

```
## 0.7741699 0.4852261
```

```
##
```

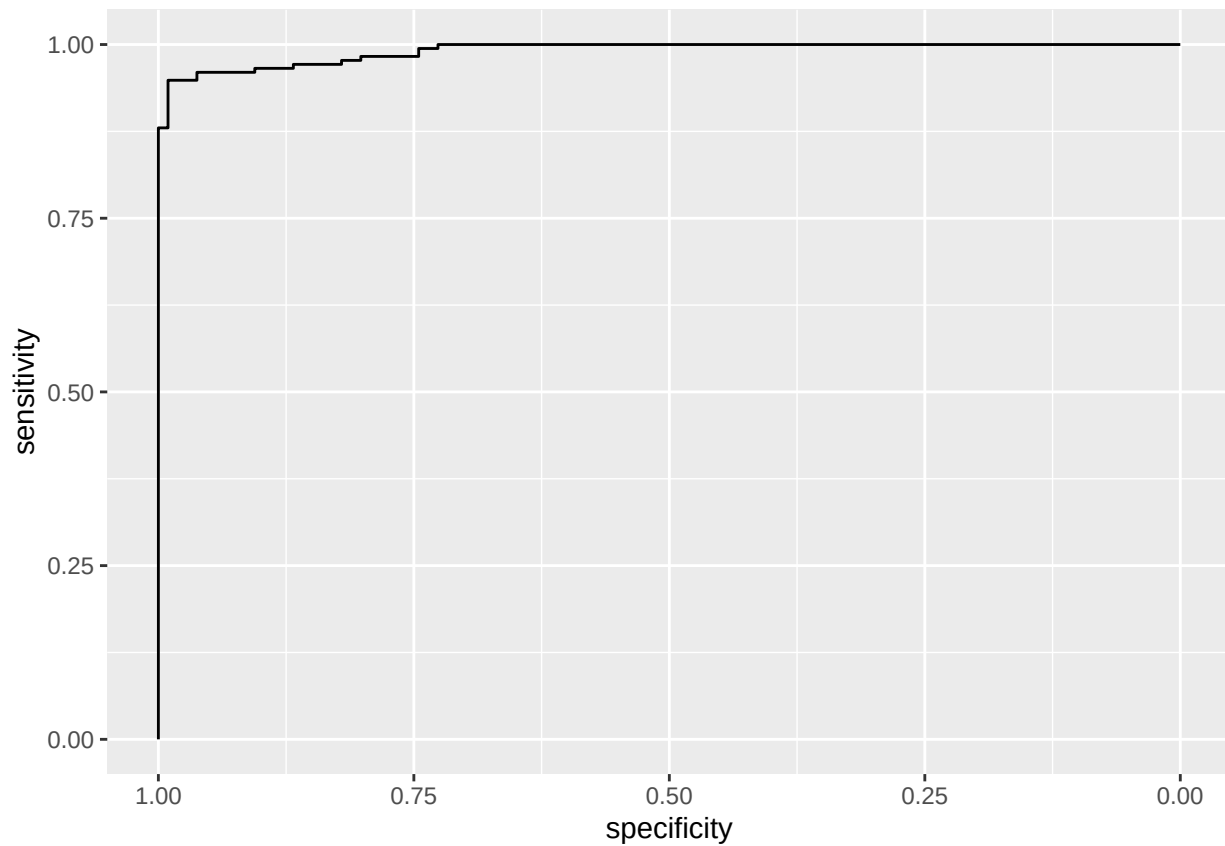
```
## Tuning parameter 'fL' was held constant at a value of 1
```

```
## Tuning
```

```
## parameter 'usekernel' was held constant at a value of TRUE
## Tuning
## parameter 'adjust' was held constant at a value of 1

## Setting levels: control = no_injury, case = injury

## Setting direction: controls < cases
```



```
## Area under the curve: 0.991
```

Comparison

Below are the models listed in order of accuracy from largest to smallest.

- Logistic regression (1)
- Penalized models (1)
- SVM (1)
- Classification trees (1)
- Random Forest (1)
- LDA (0.996)
- Neural Networks (0.996)
- QDA (0.943)
- KNN (0.822)

- Naive Bayes (0.774)

Below are the models listed in order of area under ROC curves from largest to smallest.

- Logistic regression (1)
- Penalized models (1)
- SVM (1)
- Classification trees (1)
- Random Forest (1)
- LDA (1)
- Neural Networks (1)
- QDA (1)
- Naive Bayes (0.991)
- KNN (0.966)

Recommendation

As seen above, all the models resulted in similar levels of accuracy and area under the ROC curve. Therefore, I would choose logistic regression as it is the most interpretable with the same level of performance as the more complex models.